

Frozen, Not Fine-tuned: A Controlled Evaluation of Geospatial Foundation Models for Agricultural Field Mapping

Anonymous authors

Paper under double-blind review

Abstract

Most published evaluations of geospatial foundation models (GeoFMs) fine-tune them end-to-end. For agricultural field-boundary mapping from Sentinel-2 this is unnecessary. Across six countries, a frozen GeoFM backbone with a small decoder matches or exceeds full fine-tuning, in-region for both Prithvi-EO-2.0 and TerraMind and cross-region for Prithvi-EO-2.0, while training about $110\times$ fewer parameters. Fine-tuning also reduces cross-region transfer. Two evaluation choices have hidden this result. First, common benchmarks substitute a coarse land-cover proxy (ESA WorldCover “cropland”) for true field boundaries. Second, they compare GeoFMs against a per-pixel spectral baseline. Both choices inflate the apparent GeoFM advantage. On identical pixels, a supervised U-Net trained from scratch on true labels reaches 0.89 to 0.98 AUROC in every region and matches frozen GeoFMs in five of six. India is the one exception, where the GeoFM leads. The PANGAEA benchmark shows the same pattern on its true-boundary task. The rule breaks under extreme label sparsity (around 0.1% positive pixels in Kenya); we report the boundary case and a ten-point evaluation protocol.

1 Introduction

Two routes exist for adapting a pretrained geospatial foundation model (GeoFM) to a new task: fine-tune the backbone end-to-end on target labels, or freeze it and train a small head. Most GeoFM papers fine-tune. For agricultural field-boundary mapping, that default is wrong.

Two conveniences shape most benchmarks in this area. The first is a land-cover product (ESA WorldCover “cropland” at 10 m) used as ground truth because true field boundaries are scarce. The second is a per-pixel spectral baseline (a random forest on band values) used as the non-FM comparison. This paper asks how those choices affect model ranking, and what survives once they are corrected.

Section 4 identifies two evaluation choices that bias these benchmarks; PANGAEA (Marsocci et al., 2024) shows both effects independently. The first is the label choice. Swapping true field boundaries for the WorldCover proxy raises a per-pixel random forest from near chance to GeoFM-competitive on identical pixels and reverses the apparent size bias. The second is baseline strength. A supervised U-Net trained from scratch on true labels reaches 0.89 to 0.98 AUROC in every region and matches frozen GeoFMs in five of six (India excepted). PANGAEA reports the same pattern: on AI4SmallFarms, a from-scratch U-Net beats every evaluated GeoFM by about 20 mIoU, while a GeoFM’s headline win is measured against a proxy task.

Section 5 compares frozen-backbone GeoFM features (decoder trained on frozen features) against full end-to-end fine-tuning in two settings. In-region, across six regions and two GeoFMs, the frozen backbone matches or exceeds full fine-tuning within ± 0.013 AUROC on five of six regions. Cross-region (train on one country, test on another with no target-region labels), frozen features transfer at least as well as a from-scratch U-Net and substantially better than fine-tuning, which overfits the source. The conclusion is the same in both settings: freeze the backbone, train a decoder, do not fine-tune. Section 6 reports the one regime where the

rule breaks (Kenya, around 0.1% field-pixel coverage). Section 7 quantifies the parameter savings. Section 8 gives a ten-point protocol.

Contributions. (1) A same-pixel controlled apparatus that separates the effect of label choice from that of baseline strength on GeoFM evaluation, applied across six countries with Fields-of-The-World boundaries. (2) Evidence that those two choices jointly account for the reported GeoFM advantage on field-boundary tasks, corroborated by PANGAEA. (3) An in-region and cross-region finding that fine-tuning does not improve over frozen features for field mapping, with one boundary condition (extreme label sparsity) and a roughly 110 $\times$ cost gap. (4) A ten-point evaluation protocol and a released, audited artifact.

2 Related Work

Geospatial foundation models. Recent GeoFMs pretrain on large unlabeled satellite archives and are reused across tasks: masked-autoencoder models (SatMAE (Cong et al., 2022), Scale-MAE (Reed et al., 2023), Prithvi-EO (Szwarcman et al., 2024)), contrastive and multimodal models (CROMA (Fuller et al., 2023), Clay (Clay Foundation, 2024), TerraMind (Jakubik et al., 2025), AnySat (Astruc et al., 2024)), wavelength-conditioned models (DOFA (Xiong et al., 2024)), spectral models (SpectralGPT (Hong et al., 2024)), pixel-timeseries models (Presto (Tseng et al., 2023)), and large supervised pretraining (SatlasPretrain (Bastani et al., 2023)). Their downstream evaluations almost always fine-tune the backbone. We question that default for field mapping.

Frozen features vs. fine-tuning. Outside Earth observation, a substantial literature documents that frozen pretrained features with a light head are often competitive with full fine-tuning, and that fine-tuning can harm out-of-distribution performance. Kumar et al. (2022) show that fine-tuning distorts pretrained features and underperforms frozen linear probing under distribution shift, which is the pattern we observe across regions. Locked-feature training (Zhai et al., 2022), transfer studies (Kornblith et al., 2019; Kolesnikov et al., 2020), and the linear-probe evaluation of self-supervised models (He et al., 2022) all point in the same direction. We establish the same pattern for geospatial field-boundary mapping, in and out of distribution, with an explicit boundary condition and a cost analysis.

EO benchmarks and evaluation. GEO-Bench (Lacoste et al., 2023) and PANGAEA (Marsocci et al., 2024) standardize GeoFM evaluation; we use PANGAEA’s results as external corroboration. Fields of The World (Kerner et al., 2024) supplies the true field boundaries that let us replace land-cover proxies. We audit the evaluation choices (labels and baselines) rather than propose a new model.

3 Setup

Data. We use Sentinel-2 L2A chips paired with true field boundaries from Fields of The World (Kerner et al., 2024) across six countries spanning smallholder to industrial agriculture: India, Cambodia, Vietnam, Kenya, France, and the Netherlands. For each chip we form two labels on the same pixels. The TRUE label is 1 when the pixel lies inside an FTW field polygon, rasterized to the chip grid in the chip’s own CRS. The PROXY label is 1 when the pixel is ESA WorldCover “cropland” (class 40, (Zanaga et al., 2022)). Figure 1 illustrates the contrast.

Models. Non-FM baselines: a per-pixel random forest on the 12 raw S2 bands, a 5×5 windowed random forest (300-d), and a from-scratch U-Net (Ronneberger et al., 2015) (31M parameters). GeoFMs: Prithvi-EO-2.0-300M (Szwarcman et al., 2024) and TerraMind (Jakubik et al., 2025) (with Clay and AnySat for breadth), used in two modes. Frozen: features extracted once, then a linear probe or a trained decoder. Fine-tuned: backbone trained end-to-end with the same decoder.

Protocol. All models are scored on the same held-out test pixels via a chip-grouped split (no chip in both train and test; overlap = 0; seed 20260514, 25% test). We report AUROC, mean $\pm$ std over three training



Figure 1: **The same pixels under two labels** (a Vietnam chip). Left: Sentinel-2 RGB. Middle: TRUE FTW field membership (precise parcels). Right: WorldCover “cropland” PROXY (a coarse blob). Benchmarking against the proxy quietly replaces “is this pixel in a field?” (spatial) with “is this pixel crop-like?” (spectral), which a per-pixel baseline can solve.

Table 1: **Two evaluation flaws (true-label AUROC; mean over 3 seeds).** The WorldCover proxy lifts the per-pixel RF from 0.55–0.82 (TRUE) to 0.79–0.96 (PROXY), making it foundation-model-competitive. A from-scratch U-Net on TRUE labels reaches 0.89–0.98 everywhere and matches frozen GeoFMs in five of six regions (India excepted), so the reported “GeoFM $\gg$ baseline” gap mostly reflects the weak per-pixel baseline. “frozen FM” = best frozen GeoFM with a linear probe; “FT FM” = fine-tuned Prithvi. U-Net and FT FM are mean $\pm$ std over 3 seeds; RF and the linear probe are deterministic given the split. The frozen-backbone-plus-decoder comparison is in Table 2.

Region	RF TRUE	RF PROXY	U-Net	frozen FM	FT FM
India	0.574	0.786	0.909 $\pm$.034	0.983	0.983 $\pm$.003
Cambodia	0.550	0.900	0.948 $\pm$.001	0.924	0.949 $\pm$.000
Vietnam	0.643	0.911	0.962 $\pm$.003	0.929	0.951 $\pm$.000
Kenya	0.651	0.806	0.887 $\pm$.002	0.853	0.763 $\pm$.012
France	0.695	0.959	0.977 $\pm$.000	0.961	0.978 $\pm$.001
Netherlands	0.815	0.928	0.981 $\pm$.002	0.961	0.959 $\pm$.005

seeds. The apparatus, all metric artifacts, and reproduction commands are released; every headline number was independently recomputed from raw rasters and reproduced to three decimals (Appendix A).

4 Two Evaluation Flaws

Table 1 reports, per region, the per-pixel RF and the strong models under both labels (TRUE-label AUROC unless noted).

Flaw 1: proxy labels inflate weak baselines. The proxy raises the per-pixel RF by +0.11 to +0.35 AUROC in every region, turning a near-chance field-membership predictor into one that rivals GeoFMs. The reason is that WorldCover “cropland” is spectrally separable while true field membership is a spatial property. FTW-labeled fields cover a small fraction of a scene, while WorldCover calls a majority of it cropland. The proxy therefore reframes the task as “is this managed land?” rather than “is this pixel in a field?” (Appendix).

Flaw 2: weak baselines exaggerate the GeoFM advantage. On TRUE labels a supervised U-Net reaches 0.89–0.98 AUROC in all six regions and matches frozen GeoFMs in five of them. India is the

Table 2: **Fine-tuning does not help (true-label AUROC, mean $\pm$ std, 3 seeds).** *Left, in-region* : frozen backbone + trained decoder vs. full fine-tuning of the same architecture (Prithvi). Frozen matches or beats fine-tuning on five of six regions; only Kenya (extreme sparsity) favors fine-tuning. *Right, cross-region*: mean over 10 directed transfers (train country A, test country B); frozen features transfer as well as a from-scratch U-Net and far better than fine-tuning, which overfits the source.

In-region: frozen vs. fine-tune (Prithvi)				Cross-region (10 transfers)	
Region	frozen-bb	full-FT	Δ	Model	mean AUROC
India	0.983	0.983	+0.000	frozen Prithvi	0.831
Cambodia	0.947	0.949	−0.002	frozen TerraMind	0.828
Vietnam	0.954	0.951	+0.003	U-Net (scratch)	0.825
France	0.982	0.978	+0.004	fine-tuned Prithvi	0.777
Netherlands	0.972	0.959	+0.013	spectral RF	~0.50
Kenya	0.636	0.763	−0.127		

exception: the GeoFM leads by about 0.07. The GeoFM advantage over the per-pixel RF is therefore largely an artifact of the baseline. A per-pixel model cannot use spatial context, and a 5×5 windowed RF closes only about 0.01 of the gap (Appendix). In-region, GeoFMs beat the U-Net in only one of six regions (India), and that does not generalize: Cambodia, with smaller fields than India, is a tie. The India win tracks the only high-variance U-Net in the study (± 0.034), not field size.

External corroboration (PANGAEA). On AI4SmallFarms (Persello et al., 2023) (manually digitized field boundaries), PANGAEA (Marsocci et al., 2024) reports a from-scratch U-Net at 46.3 mIoU, beating every evaluated GeoFM (Prithvi 26.9, DOFA 27.1, Scale-MAE 21.5). Conversely, the Prithvi-EO-2.0 headline of +8.1 mIoU over a from-scratch U-Net on multi-temporal crop segmentation (Szwarcman et al., 2024) is measured against USDA Cropland Data Layer proxy labels. Both flaws appear in the published literature, independent of our apparatus.

5 Fine-tuning Underperforms Frozen Features: In and Out of Distribution

Holding the decoder, recipe, labels, and evaluation pixels fixed, Table 2 compares frozen-backbone GeoFM features against full end-to-end fine-tuning.

In-region. With the backbone frozen and only a decoder trained, performance matches full fine-tuning on five of six regions (within ± 0.013 AUROC) for both Prithvi and TerraMind (TerraMind: India +0.001, France +0.006). Fine-tuning the 300M-parameter backbone gives no in-region gain. An earlier apparent fine-tuning advantage came from an architecture confound: comparing a fine-tune-plus-decoder against a frozen linear probe. Giving the frozen backbone the same decoder removes the gap.

Cross-region. Transferring across countries without target-region labels (10 directed transfers among five regions; Vietnam is omitted from the transfer subset), frozen GeoFM features (0.831 Prithvi, 0.828 TerraMind) transfer as well as a from-scratch U-Net (0.825) and better than fine-tuned Prithvi (0.777). Fine-tuning is the worst transferer in nearly every direction: it overfits the source region. Frozen features win or tie 7 of 10 directed transfers; spectral features collapse toward chance. We do not claim frozen features beat the U-Net on transfer (they tie), but they achieve it with no target-region training and at a fraction of the cost (Section 7), and they avoid fine-tuning’s degradation. Both settings give the same conclusion: do not fine-tune the backbone.

6 When the Rule Breaks: Characterizing the Boundary

Kenya is the one exception in both experiments: in-region, full fine-tuning beats the frozen backbone ($\Delta = -0.127$), and cross-region the U-Net wins. Three properties distinguish Kenya. First, label sparsity: Kenya’s true-field-pixel coverage in training masks is the lowest of any region at about 0.1%, versus India 0.3%,

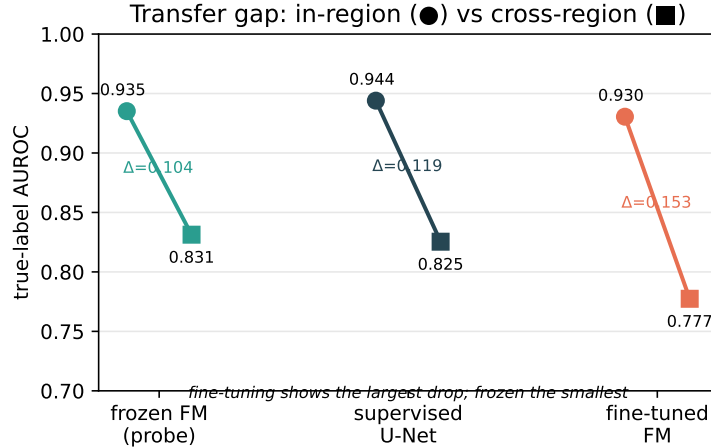


Figure 2: **Fine-tuning transfers worst.** In-region (●) vs. cross-region (■) mean AUROC per model class. The fine-tuned GeoFM has the largest in-region-to-cross-region drop ($\Delta=0.15$); frozen features have the smallest ($\Delta=0.10$); the U-Net sits between the two.

Table 3: **Cost asymmetry.** Trainable parameters and per-region training cost. Frozen options match full fine-tuning or the U-Net at a small fraction of the trainable parameters.

Method	trainable params	per-region train	target labels
per-pixel RF	0 (non-parametric)	seconds, CPU	yes
frozen FM + linear probe	$\sim 1,000$	seconds, CPU	yes (source only, transfer)
frozen backbone + decoder	2.75M	~ 12 GPU-min	yes
supervised U-Net	31M	~ 15 GPU-min	yes
full fine-tune (Prithvi)	307M	~ 20 GPU-min	yes

Netherlands 3.8%, France 11%, Vietnam 18%, Cambodia 23%. This is not field size: Cambodia has the smallest fields (79% under 0.3 ha) but the highest coverage (23%) and obeys the rule, so the boundary tracks annotated-positive density, not parcel size. Second, training-set size: Kenya has the fewest chips (155). Third, spatial fragmentation: Kenya’s positive pixels form scattered micro-clusters rather than contiguous parcels, which leaves the decoder with far fewer within-field-interior contexts per chip than Cambodia or France. Under this combination, frozen-feature decoders are unstable (seed std up to ± 0.04), and an end-to-end CNN is the more robust choice. Threshold metrics sharpen the boundary: at probability threshold 0.5, both the frozen probe and full fine-tune collapse to all-negative on Kenya ($F1 = IoU = 0$; Appendix Table 8); only the supervised U-Net recovers any positives ($F1 = 0.115$). Kenya is a boundary case across both rank and threshold metrics, not an AUROC artifact.

We do not state a hard numeric threshold. With one region below about 0.2% coverage, the transition is an $n=1$ observation, and a precise cutoff would overclaim. A working diagnostic instead: when annotated positives are very sparse ($\lesssim 0.2\%$ of pixels) and the training set is small, expect frozen-feature decoders to become unstable, and prefer a from-scratch supervised CNN. Practitioners can check their own positive rate and seed variance to decide. Establishing the exact boundary is future work.

7 Cost-Efficiency of Frozen Features

Table 3 contrasts trainable parameters and per-region training cost. A frozen backbone with a decoder trains 2.75M parameters versus 307M for full fine-tuning, about $110\times$ fewer, while matching its accuracy. A frozen linear probe trains about 10^3 parameters in seconds on CPU and matches a 31M-parameter U-Net trained on GPU. For cross-region deployment, the frozen probe needs no target-region labels.

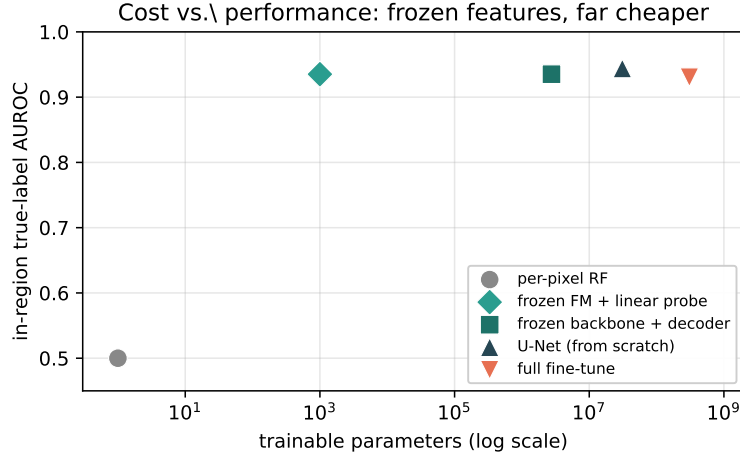


Figure 3: **Cost vs. performance.** In-region AUROC against trainable parameters (log scale). Frozen options (a $\sim 1\text{K}$ -parameter linear probe, or a 2.75M-parameter decoder) sit at the cheap, high-accuracy corner, matching the 31M U-Net and the 307M full fine-tune.

8 A Corrected-Evaluation Protocol

We distill the study into a ten-point checklist for field-level GeoFM benchmarking. **Labels:** (1) evaluate against true field boundaries, not a land-cover proxy; treat proxy-only rankings as provisional. (2) Make the label comparison same-pixel paired. **Baselines:** (3) include a strong task-specific supervised baseline (U-Net), not only a per-pixel spectral model. (4) Report a frozen-feature probe and a fine-tuned variant; do not assume fine-tuning helps. **Splits:** (5) group by chip. (6) Add a scene or tile-disjoint split, and a cross-region split. **Preprocessing:** (7) use each model’s own normalization; exclude no-data. **Statistics:** (8) cluster uncertainty at the chip level, use multiple seeds, apply multiple-comparison correction. **Interpretation:** (9) run an edge-vs-interior control before claiming boundary understanding. (10) Report across field-size and label-sparsity regimes, and report trainable-parameter and compute cost alongside accuracy.

Worked example: re-reading a published ranking. The Prithvi-EO-2.0 report (Szwarcman et al., 2024) headlines a +8.1 mIoU win for the foundation model over a from-scratch U-Net on multi-temporal crop segmentation. Protocol step (1) flags it as provisional: the labels come from the USDA Cropland Data Layer, a per-pixel land-cover proxy rather than true field boundaries, so the metric rewards reproducing the proxy’s per-pixel artifacts (Flaw 1). With true boundaries and a strong baseline (steps 1 to 3), PANGAEA (Marsocci et al., 2024) reports on AI4SmallFarms (Persello et al., 2023) (manually digitized parcels) a from-scratch U-Net at 46.3 mIoU, beating every evaluated GeoFM (Prithvi 26.9, DOFA 27.1, Scale-MAE 21.5). Applying the protocol inverts the takeaway one would draw from the proxy-labeled headline, using only published numbers.

Practitioner implications. (i) Cost: a frozen backbone with a decoder trains about $110\times$ fewer parameters than full fine-tuning, and a linear probe about 10^3 (Table 3). It is cheaper to train, store, and version. (ii) Label-scarce deployment: for a new region with no local labels, a frozen-feature model trained elsewhere transfers as well as a region-specific U-Net and better than a fine-tuned GeoFM. Fine-tuning should therefore be avoided precisely when target labels are absent. (iii) Stability: frozen pipelines show lower seed variance outside the extreme-sparsity regime. The default for agricultural field mapping should be to extract frozen GeoFM features once, train a small decoder, and skip fine-tuning unless positives are very sparse.

9 Discussion, Limitations, Conclusion

Scope. Our claims are scoped to agricultural field-boundary mapping from single-date Sentinel-2 with true field labels. They do not necessarily extend to other tasks (e.g. crop-type, change detection), other sensors, or multi-temporal regimes. GeoFM pretraining may help there.

Why not “GeoFMs are useless.” We do not argue against GeoFMs. We argue that current evaluations overstate their advantage for in-distribution field mapping with reasonable supervision, and that their practical value here is as frozen feature extractors. They are cheap, stable, and transfer-competitive in that configuration, rather than as backbones to fine-tune.

Limitations. Four GeoFMs evaluated here (PANGAEA covers more, with the same pattern); a single supervised architecture (PANGAEA independently replicates with its own U-Net); FTW partial annotation in sparse regions (our sensitivity analysis shows the conclusions are stable, Appendix); and the cross-region tie rather than a win.

Connection to the broader foundation-model literature. The cross-region result empirically demonstrates, for geospatial models, what Kumar et al. (2022) established theoretically and for vision: fine-tuning can distort pretrained features and underperform frozen probing under distribution shift. The pattern recurs across modality, domain, and shift type: multispectral satellite imagery rather than natural images; field-boundary mapping rather than classification; geographic rather than corruption-based distribution shift. This suggests the “freeze, don’t fine-tune” recommendation may extend to other Earth-observation tasks where labeled supervision is regionally concentrated. Our apparatus is designed to test that hypothesis going forward.

Conclusion. For agricultural field mapping from Sentinel-2, the recipe is: evaluate on true field boundaries, compare against a strong task-specific supervised baseline, and deploy frozen GeoFM features rather than fine-tune the backbone. Two evaluation choices, proxy labels and weak per-pixel baselines, have made fine-tuned GeoFMs appear to dominate. The same-pixel controlled apparatus, corroborated by PANGAEA, shows that a from-scratch supervised U-Net matches them in all but one region (India) once both flaws are corrected. Across six regions, two foundation models, and both in- and out-of-distribution settings, fine-tuning the backbone does not help, and it hurts cross-region transfer. To our knowledge, this is the first controlled multi-region demonstration of the result for geospatial field mapping. It holds at roughly $110\times$ lower trainable-parameter cost. We release the apparatus, the ten-point protocol, and all artifacts so that future GeoFM evaluations can avoid both flaws by construction.

References

- Guillaume Astruc et al. AnySat: An earth observation model for any resolutions, scales, and modalities. *arXiv preprint arXiv:2412.14123*, 2024.
- Fayyen Bastani et al. SatlasPretrain: A large-scale dataset for remote sensing image understanding. In *International Conference on Computer Vision (ICCV)*, 2023.
- Clay Foundation. Clay foundation model: An open source ai model for earth. <https://clay-foundation.github.io/model/>, 2024. v1.5; open community model.
- Yezhen Cong et al. SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Anthony Fuller, Koreen Millard, and James R. Green. CROMA: Remote sensing representations with contrastive radar-optical masked autoencoders. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

- Danfeng Hong et al. SpectralGPT: Spectral remote sensing foundation model. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.
- Johannes Jakubik et al. TerraMind: Large-scale generative multimodality for earth observation. In *International Conference on Computer Vision (ICCV)*, 2025. arXiv:2504.11171.
- Hannah Kerner et al. Fields of the world: A machine learning benchmark dataset for global agricultural field boundary segmentation. *arXiv preprint arXiv:2409.16252*, 2024.
- Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Joan Puigcerver, Jessica Yung, Sylvain Gelly, and Neil Houlsby. Big transfer (BiT): General visual representation learning. In *European Conference on Computer Vision (ECCV)*, 2020.
- Simon Kornblith, Jonathon Shlens, and Quoc V. Le. Do better ImageNet models transfer better? In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations (ICLR)*, 2022.
- Alexandre Lacoste et al. GEO-Bench: Toward foundation models for earth monitoring. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023. arXiv:2306.03831.
- Valerio Marsocci et al. PANGAEA: A global and inclusive benchmark for geospatial foundation models. *arXiv preprint arXiv:2412.04204*, 2024.
- Claudio Persello et al. AI4SmallFarms: A dataset for crop field delineation in Southeast Asian smallholder farms. *IEEE Geoscience and Remote Sensing Letters*, 20, 2023.
- Colorado J. Reed et al. Scale-MAE: A scale-aware masked autoencoder for multiscale geospatial representation learning. In *International Conference on Computer Vision (ICCV)*, 2023.
- Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015.
- Daniela Szwarcman et al. Prithvi-EO-2.0: A versatile multi-temporal foundation model for earth observation applications. *arXiv preprint arXiv:2412.02732*, 2024.
- Gabriel Tseng, Ruben Cartuyvels, Ivan Zvonkov, Mirali Purohit, David Rolnick, and Hannah Kerner. Lightweight, pre-trained transformers for remote sensing timeseries (Presto). *arXiv preprint arXiv:2304.14065*, 2023.
- Zhitong Xiong et al. Neural plasticity-inspired multimodal foundation model for earth observation (DOFA). *arXiv preprint arXiv:2403.15356*, 2024.
- Daniele Zanaga, Ruben Van De Kerchove, et al. ESA WorldCover 10m 2021 v200. *Zenodo*, 2022. doi: 10.5281/zenodo.7254221.
- Xiaohua Zhai, Xiao Wang, Basil Mustafa, Andreas Steiner, Daniel Keysers, Alexander Kolesnikov, and Lucas Beyer. LiT: Zero-shot transfer with locked-image text tuning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

A Reproducibility and Controls

All experiments use the same canonical split (chip-grouped, seed 20260514, overlap 0). Headline numbers were independently recomputed from raw rasters and features and reproduced to three decimals; label-to-imagery alignment was verified at 100% (every positive pixel inside a field polygon, every negative outside) in all six regions. Controls: a per-pixel ablation (1×1 convolutions) collapses to $\approx$ RF, and a 5×5 windowed

RF gains only about 0.01 AUROC, which confirms that the U-Net and GeoFM advantage is learned spatial context rather than naive pooling. An edge-pixel control localizes the advantage to within-field interior context. An FTW partial-labeling sensitivity test (restricting negatives to high-confidence non-cropland) leaves the GeoFM true-task AUROC almost unchanged (e.g. India Prithvi $0.983 \rightarrow 0.987$). Full cross-region transfer (all 10 directed pairs, three frozen GeoFMs) and per-seed tables are released with the artifact. Code, the controlled-evaluation apparatus, and all metric artifacts (with checksums and end-to-end reproduction commands) are available at <https://anonymous.4open.science/r/frozen-not-finetuned-eval> (anonymized for review; a Zenodo DOI and Croissant metadata will accompany the camera-ready).

Table 4: **Training recipe, held fixed across compared models.** All supervised models share loss, augmentation, optimizer family, and schedule; only the backbone treatment (frozen vs. fine-tuned vs. from-scratch) differs. Evaluation pixels and the chip-grouped split are identical for every model and every seed.

Component	Setting
Loss	BCE (capped positive weight, ≤ 50) + Dice
Optimizer	U-Net: Adam, lr 10^{-3} ; GeoFM: AdamW (backbone 10^{-4} , decoder 10^{-3})
LR schedule	linear warmup (5 epochs, start factor 0.1) + cosine decay
Augmentation	random horizontal/vertical flips
Input	U-Net: native chip, 12 S2 bands (reflect-padded); GeoFM: 224×224
Decoder	GeoFM: shared 4-stage bilinear-upsample conv decoder (identical for frozen & fine-tuned); U-Net: 4-level (31M)
Epochs	U-Net 150 (from scratch); GeoFM fine-tune / frozen-probe 80
Selection	last epoch (no early stopping); mean $\pm$ std over 3 training seeds

Table 5: **Full cross-region transfer** (true-label AUROC, mean over 3 seeds; train country A, test country B’s canonical test pixels; cross-region transfer covers a five-region subset, Vietnam omitted). Frozen GeoFM features are best or tied-best in 7 of 10 directed transfers; fine-tuning (Prithvi-ft) is the worst transferer in nearly all; the U-Net wins the three Kenya-involved transfers (the extreme-sparsity boundary). The one direction in which fine-tuning helps Prithvi’s transfer is Cambodia $\rightarrow$ India (Prithvi-ft 0.859 > frozen Prithvi 0.832); frozen TerraMind (0.879) is nonetheless highest there, so the “don’t fine-tune” conclusion still holds at the model-class level.

A $\rightarrow$ B	U-Net	Prithvi-ft	frozen-Prithvi	frozen-TerraMind
India $\rightarrow$ Cambodia	0.691	0.671	0.796	0.849
Cambodia $\rightarrow$ India	0.836	0.859	0.832	0.879
France $\rightarrow$ Netherlands	0.938	0.906	0.943	0.927
Netherlands $\rightarrow$ France	0.938	0.909	0.946	0.925
India $\rightarrow$ France	0.796	0.721	0.850	0.848
France $\rightarrow$ India	0.891	0.868	0.913	0.911
India $\rightarrow$ Kenya	0.690	0.576	0.721	0.748
Kenya $\rightarrow$ India	0.748	0.706	0.704	0.669
Cambodia $\rightarrow$ Kenya	0.883	0.839	0.853	0.808
Kenya $\rightarrow$ Cambodia	0.843	0.719	0.756	0.717
mean	0.825	0.777	0.831	0.828

Table 6: **Spatial-pooling control (true-label AUROC)**. A 5×5 windowed random forest (300-d) gains only about 0.01 over the per-pixel RF and stays far below the best foundation model. This confirms that the GeoFM and U-Net advantage is learned spatial context, not naive local pooling. A per-pixel U-Net ablation (1×1 convolutions, no spatial context) likewise collapses to $\approx$ RF (India 0.623, Vietnam 0.678).

Region	per-pixel RF	windowed RF (5×5)	best FM
India	0.574	0.582	0.983
Vietnam	0.643	0.657	0.929
Kenya	0.651	0.641	0.853
France	0.695	0.703	0.961
Netherlands	0.815	0.827	0.961

Table 7: **FTW partial-labeling sensitivity (true-label AUROC)**. Restricting negatives to high-confidence non-cropland (dropping ambiguous, possibly-unlabeled field pixels) leaves the frozen-GeoFM score barely changed, while the spectral RF rises. The GeoFM result is therefore not an artifact of FTW’s partial annotation: the RF’s errors were concentrated on crop-like non-field pixels. The large France RF gain ($0.695 \rightarrow 0.947$) reflects that many of its true-negatives are spectrally crop-like (pasture, vineyards) and inseparable by a per-pixel RF. Removing them leaves an easy spectral task. The GeoFM was already near-ceiling, so it barely moves.

Region	RF (all neg)	RF (high-conf)	Prithvi (all neg)	Prithvi (high-conf)
India	0.574	0.665	0.983	0.987
France	0.695	0.947	0.961	0.968
Kenya	0.651	0.698	0.766	0.774

Table 8: **Threshold-based metrics complement AUROC** (single seed, seed 0; F1 and IoU at probability threshold 0.5; AP is threshold-free). The AUROC story holds across rank- and threshold-based metrics in dense-positive regions (Cambodia, Vietnam, France, Netherlands), where U-Net, frozen probe, and full fine-tune cluster within about 0.02 to 0.05 IoU. India keeps its single-region FM lead on AUROC and AP; F1 and IoU are close across models there because positives are sparse (about 0.3%). Kenya (about 0.1% positives) shows the sharpest gap: both the frozen probe and the full fine-tune collapse to F1=IoU=0 at the standard threshold, and AP drops to 0.32 to 0.38, while the supervised U-Net retains AP=0.58 and F1=0.115. The supervised CNN is therefore the more robust choice in the extreme-sparsity regime, across every metric and not only AUROC. “Frozen” = Prithvi backbone frozen, decoder trained (apples-to-apples vs. full fine-tune).

Region	Model	AUROC	AP	F1	IoU
India	U-Net	0.960	0.893	0.583	0.412
India	Frozen	0.985	0.972	0.573	0.401
India	Full-FT	0.981	0.971	0.617	0.446
Cambodia	U-Net	0.951	0.927	0.914	0.842
Cambodia	Frozen	0.948	0.913	0.918	0.848
Cambodia	Full-FT	0.942	0.904	0.916	0.846
Vietnam	U-Net	0.957	0.938	0.916	0.845
Vietnam	Frozen	0.954	0.927	0.913	0.839
Vietnam	Full-FT	0.951	0.920	0.907	0.831
Kenya	U-Net	0.891	0.583	0.115	0.061
Kenya	Frozen	0.667	0.316	0.000	0.000
Kenya	Full-FT	0.757	0.375	0.000	0.000
France	U-Net	0.977	0.962	0.942	0.890
France	Frozen	0.982	0.972	0.921	0.853
France	Full-FT	0.978	0.971	0.906	0.829
Netherlands	U-Net	0.982	0.980	0.892	0.805
Netherlands	Frozen	0.971	0.966	0.818	0.692
Netherlands	Full-FT	0.950	0.953	0.788	0.650